

Static Cosmology

Lecture 11: Astronomy of Medium Condensates

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Overview

Compact medium condensates are inferred from orbital, optical, and kinematic likelihoods with a selection function.

1 Observational setup

Source coordinates, observing band, selection function, and calibration state are fixed at the outset.

$$M_c = 1.9M_\odot$$

2 Transport or equilibrium equation

The governing equation is solved first in a homogeneous limit and then with the stated spatial dependence.

$$p(M|D) \propto L_{orbit}L_{opt}L_{kin}p(M)$$

3 Connection to data

Each model quantity is tied to a measured flux, spectrum, angle, count, or temperature.

$$\varphi(M) = selection^{-1}dN/dM$$

4 Degeneracies

Calibration, population, and medium parameters are varied separately to expose their degeneracies.

5 Example

The worked example carries units and uncertainty from the input catalog to the reported result.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Jeong, ch. 11
- Dense Object Catalog DOC-26
- CMT D block